

# Tracking How Consent Interfaces Evolve

A repeatable, evidence-linked framework for comparing the same website over time

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RQ1  
repeatable measure

RQ2  
matched versions

RESULT  
validated trajectory

improved | regressed | mixed | stable | insufficient

THESIS RQ1 defines the ruler. RQ2 repeats the measurement. Evidence review determines whether a change has direction.

## 01 QUESTION + MEASURE

A snapshot can describe one consent interface. It cannot show how that interface evolves.

The study compares repeated observations of the same site and asks whether user choice became easier, clearer, more balanced, or worse.

### RQ1: the repeatable ruler

#### Path availability

Can users reach Reject and Customize?

#### Path effort

How much interaction does each path require?

#### Transparency

What is disclosed, and how clearly?

#### Unbiased choice

Are options visually and structurally balanced?

### Directional coding

|              |                                                 |
|--------------|-------------------------------------------------|
| IMPROVED     | one or more dimensions improve; none regress    |
| REGRESSED    | one or more regress; none improve               |
| MIXED        | dimensions move in both directions              |
| STABLE       | no meaningful component change                  |
| INSUFFICIENT | context, evidence, or scoring is not comparable |

#### Guardrail

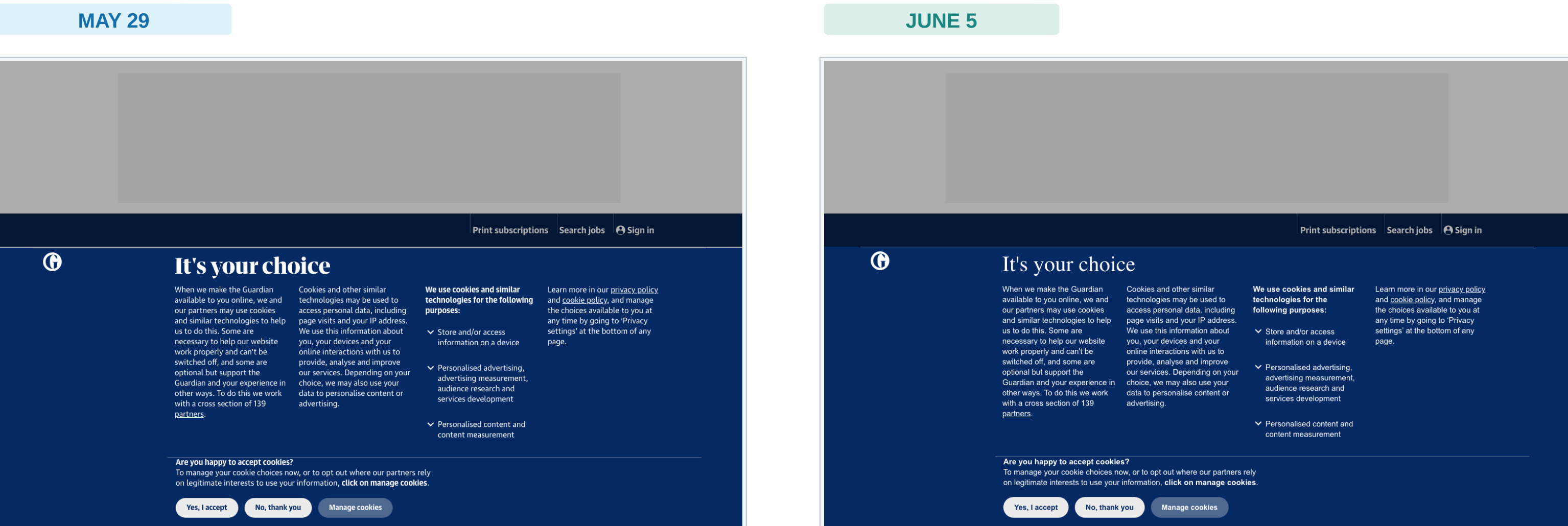
Copy, DOM, layout, and image hashes trigger review. They do not show improvement or regression by themselves.

## 02 MATCHED LONGITUDINAL DESIGN

Same site + same context + same ruler

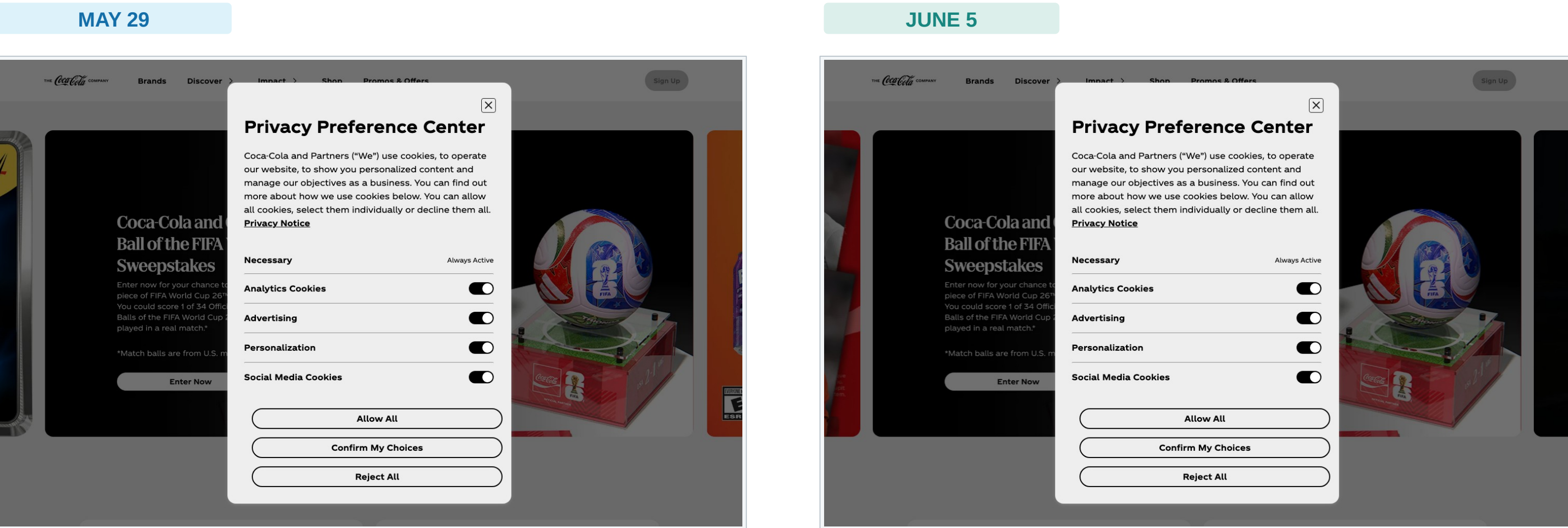


### Matched pair 1: The Guardian



Raw alert: D | Evidence review: same visible choice structure | Direction: insufficient evidence

### Matched pair 2: Coca-Cola



Stored controls look stable; automated tier changed; a later detector fix changed recognition. Back-coding is required.

## 03 PILOT RESULT + NEXT STUDY

5

sites

1

matched interval

5

raw candidates

### CURRENT FINDING

The pipeline can surface evidence-linked change candidates. The current one-interval pilot does not establish improvement or regression for any site.

### Three context cases

CNN, Booking.com, and NerdWallet show no visible first-screen banner at either stored time point. They are repeated non-observations, not banner-path failures.

### Why the depth is not enough yet

- one validated interval, not a long trajectory
- capture context is not fully modeled
- scorer-version confound in Coca-Cola
- no reliability estimate
- five selective cases

### Next study protocol

- 1 Collect at least three validated time points per site
- 2 Freeze geography, language, viewport, browser, and consent state
- 3 Store scorer version and back-code after detector changes
- 4 Manually validate component deltas; double-code a subset
- 5 Report trajectories only after evidence passes these gates

### CONTRIBUTION

A traceable method for separating real consent-interface evolution from technical noise.

## CONCLUSION

We know how to build the comparison. We do not yet have enough validated time to claim which sites improved.

### Methods and evidence

CONCEPTS.md | audit\_report\_summary.csv | longitudinal\_summary.csv | directional review 2026-07-29